

Exercise for Chapter 6 and 7

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Summary of chapter 6

1. Knowledge of the heat capacities and the entropies of substances and the heats of formation of compounds allows the enthalpy and entropy changes to be evaluated for any process—that is, for phase changes and chemical reactions.
2. The heat capacity of solids can be modeled by the statistical thermodynamic models of Einstein and Debye. The Debye model is best at low temperatures as it displays the experimentally observed T^3 dependence at low temperatures for non conductors.
3. Heat capacities can also be fit to experimental data via an equation of the form

$$c_P = a + bT + \frac{c}{T^2}$$

Summary of chapter 6

4. Since enthalpy does not have an absolute value, it is conventional to assign the value of zero to the enthalpy of all elements in their stable states of existence at 298 K and to consider changes in enthalpy with respect to this reference state.
5. The entropy of all substances which are in complete internal equilibrium is zero at 0 K. This is the Nernst–Planck–Simon statement of the Third Law of Thermodynamics.
6. Both the enthalpy and entropy are dependent on pressure and temperature; however, the pressure dependence of the enthalpy and entropy of condensed phases is normally small enough to be ignored, especially when the pressure of interest is in the range 0–1 atm.

Summary of chapter 6

7. The variation of enthalpies of transformation with temperature and pressure can be calculated by the application of the First Law of Thermodynamics to the changes in enthalpy with temperature and pressure, as well as the temperature dependence of the heat capacities and expansion coefficients.
8. The enthalpies and entropies of transformation at the melting point and boiling point of metals can be estimated by the rules of Richard and Trouton.

Summary of chapter 6

9. The determination of ΔH_T and ΔS_T for any change of state at any temperature and pressure allows the all-important change in the Gibbs free energy for the change of state to be calculated as

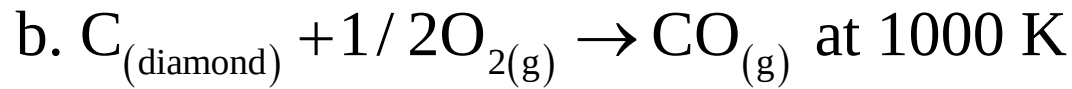
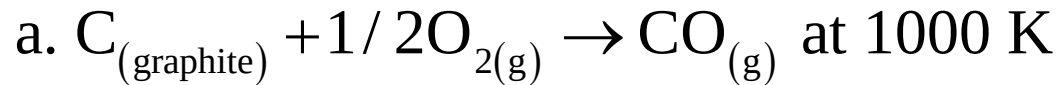
$$\Delta G_T = \Delta H_T - T \Delta S_T$$

10. Since consideration of the Gibbs free energy in any isothermal, isobaric process provides the criterion for equilibrium, the equilibrium state of a system can be determined from knowledge of the thermochemical properties of the system.

Chapter 6

放热

6.2 Which of the following two reactions is the more exothermic?



$$C_{P,C_{(\text{graphite})}} = 0.11 + 38.94 \times 10^{-3} T - 1.48 \times 10^5 T^{-2} \quad \text{J/mole} \cdot \text{K} \quad (298-1100 \text{ K})$$

$$C_{P,C_{(\text{diamond})}} = 9.12 + 13.22 \times 10^{-3} T - 6.19 \times 10^5 T^{-2} \quad \text{J/mole} \cdot \text{K} \quad (298-1200 \text{ K})$$

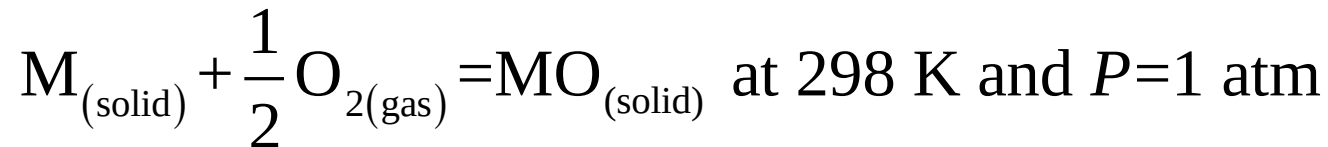
$$C_{P,O_{2(g)}} = 29.96 + 4.18 \times 10^{-3} T - 1.67 \times 10^5 T^{-2} \quad \text{J/mole} \cdot \text{K} \quad (298-3000 \text{ K})$$

$$C_{P,CO_{(g)}} = 28.41 + 4.10 \times 10^{-3} T - 0.46 \times 10^5 T^{-2} \quad \text{J/mole} \cdot \text{K} \quad (298-2500 \text{ K})$$

$$\Delta H_{298 \text{ K}}^0 (\text{graphite} \rightarrow \text{diamond}) = 1900 \quad \text{J}$$

6.4 Enthalpy as a function of temperature and composition

- ✓ Since H does not have an absolute value (only changes in H can be measured), it is convenient to introduce a convention which will allow the comparison of different enthalpy–temperature diagrams. **This convention assigns the value of zero to the enthalpy of *elements in their stable states at 298 K (25°C) and $P=1$ atm.***
- ✓ The enthalpy of a compound at 298 K and $P=1$ atm is simply the enthalpy of formation of one mole of the compound from its elements at 298 K and $P=1$ atm.



$$\Delta H_{298} = H_{\text{MO(s)298}} - H_{\text{M(s)298}} - \frac{1}{2} H_{\text{O2(gas)298}}$$

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Solution to 6.2

$$C_{P,C_{(\text{graphite})}} = 0.11 + 38.94 \times 10^{-3} T - 1.48 \times 10^5 T^{-2} - 17.38 \times 10^6 T^2$$

J/mole · K (298-1100 K)

$$\begin{aligned} H_{(\text{graphite},1000)} &= 0.11(1000 - 298) + 38.94 / 2 \times 10^{-3} (1000^2 - 298^2) \\ &\quad + 1.48 \times 10^5 (1/1000 - 1/298) - 17.38 \times 10^6 / 3 (1000^3 - 298^3) \\ &= 11830 \text{ J/mole} \end{aligned}$$

Chapter 6

$$C_{P,C(\text{diamond})} = 9.12 + 13.22 \times 10^{-3} T - 6.19 \times 10^{-5} T^{-2}$$

J/mole · K (298-1200 K)

$$\Delta H_{298 \text{ K}}^0 (\text{graphite} \rightarrow \text{diamond}) = 1900 \text{ J/mole}$$

$$\begin{aligned} H_{(\text{diamond}, 1000)} &= 1900 + 9.12(1000 - 298) + 13.22 / 2 \times 10^{-3} (1000^2 - 298^2) \\ &\quad + 6.19 \times 10^{-5} (1/1000 - 1/298) \\ &= 12867 \text{ J/mole} \end{aligned}$$

For diamond → graphite $\Delta H_{1000} = 11830 - 12867 = -1037 \text{ J/mole}$

Thus the oxidation of diamond at 1000 K is 1037 J/mole more exothermic than the oxidation of graphite.

Chapter 6

6.3 Calculate the change in enthalpy and the change in entropy at 1000 K for the reaction $\text{CaO}_{(s)} + \text{TiO}_{2(s)} \rightarrow \text{CaTiO}_{3(s)}$

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Table A.2 The Constant-Pressure Molar Heat Capacities of Various Substances ($c_p = a + bT + cT^{-2}$ J/mole·K)

Substance	a	b × 10 ³	c × 10 ⁻⁵	Range, K	Remarks
Ag	21.30	8.54	1.51	298–1234 (T_m)	
Ag(l)	30.50	—	—	1234–1600	
Al(s)	20.67	12.38	—	298–933 (T_m)	
Al(l)	31.76	—	—	933–1600	
Al ₂ O ₃	106.6	17.78	-28.53	298–2325 (T_m)	
Ba(s)	-473.2	1587.0	128.2	298–648	
Ba(g)	-5.69	80.33	—	648–1003	
BaO	53.30	4.35	-8.30	298–2286 (T_m)	
BaTiO ₃	121.46	8.54	-19.16	298–1800	
C(graphite)	0.11	38.94	-1.48	298–1100	-17.38 × 10 ⁻⁸ J ²
C ₂ (graphite)	24.43	0.44	-31.63	1100–4000	
C(diamond)	9.12	13.22	-6.19	298–1200	
CO	28.41	4.10	-0.46	298–2500	
CO ₂	44.14	9.04	-8.54	298–2500	
Ca(s)	25.37	-7.26	—	298–716	23.72 × 10 ⁻⁸ J ²
Ca(g)	-0.36	41.25	—	716–1115	
CaO	49.62	4.51	-6.95	298–1177	
CaTiO ₃	127.49	5.69	-27.99	298–1530	
Cr(s)	24.43	9.87	-3.68	298–2130 (T_m)	
Cr ₂ O ₃	119.37	9.30	-15.65	298–1800	
Cu(s)	22.64	6.28	—	298–1356 (T_m)	
Fe(s)	37.12	6.17	—	298–1183/1664–1809	
Fe(l)	24.47	8.45	—	1187–1664	
Fe(g)	41.8	—	—	1809–1873	
H ₂ O(g)	30.00	10.71	0.33	298–2500	
O ₂ (g)	29.96	4.18	-1.67	298–3000	
2MgO·2Al ₂ O ₃ ·5SiO ₂	626.34	91.21	-200.83	298–1738 (T_m)	
N ₂	27.87	4.27	—	298–2500	
Si ₃ N ₄	70.54	98.74	—	298–900	
SiO ₂ (α-quartz)	43.89	1.00	-6.02	298–847	
Ti	22.09	10.46	—	298–1155	
TiO ₂ (rutile)	75.19	1.17	-18.20	298–1800	
Zr(s)	21.97	11.63	—	298–1136	
Zr(l)	23.22	4.64	—	1136–2128	
ZrO ₂ (s)	69.62	7.53	-14.06	298–1478	
ZrO ₂ (g)	74.48	—	—	1478–2950 (T_m)	

Table A.3 The Standard Molar Heats of Formation and Molar Entropies of Various Substances at 298 K

Substance	ΔH_{298}° , J	S_{298}° , J/K
Al ₂ O ₃	-1,675,700	50.9
Ba	—	62.4
BaO	-548,100	72.1
BaTiO ₃	-1,653,100	107.9
C ₂ (graphite)	—	5.73
C ₂ (diamond)	1,900	2.43
CH ₄	-74,800	186.3
CO	-110,500	197.5
CO ₂	-393,500	213.7
Ca	—	41.6
CaO	-634,900	38.1
CaTiO ₃	-1,660,600	93.7
3CaO·Al ₂ O ₃ ·3SiO ₂	-6,646,300	241.4
CaO·Al ₂ O ₃ ·SiO ₂	-3,293,200	144.8
CaO·Al ₂ O ₃ ·2SiO ₂	-4,223,700	202.5
2CaO·Al ₂ O ₃ ·SiO ₂	-3,989,400	198.3
Cr ₂ O ₃	-1,134,700	81.2
H ₂ O(g)	-241,800	232.9
N ₂	—	191.5
O ₂	—	205.1
SiO ₂ (α-quartz)	-910,900	41.5
Si ₃ N ₄	-744,800	113.0
Ti	—	30.7
TiO	-543,000	34.7
Ti ₂ O ₃	-1,521,000	77.2
Ti ₃ O ₅	-2,459,000	129.4
TiO ₂	-944,000	50.6
Zr	—	39.0
ZrO ₂	-1,100,800	50.4

Table A.5 Molar Heats of Melting and Transformation

Substance	Trans.	ΔH_{trans} , J	T_{trans} , K
Ag	s → l	11,090	1,234
Al	s → l	10,700	934
Al ₂ O ₃	s → l	107,500	2,324
Au	s → l	12,600	1,338
Ba	α → β	630	648
Ba	β → l	7,650	1,003
Cu	s → l	12,970	1,356
Ca	α → β	900	716
CaF ₂	s → l	31,200	1,691
Fe	α → γ	670	1,187
Fe	γ → δ	840	1,664
Fe	δ → l	13,770	1,809
H ₂ O	s → l	6,008	273
K ₂ O·B ₂ O ₃	s → l	62,800	1,220
MgF ₃	s → l	58,160	1,563
Na ₂ O·B ₂ O ₃	s → l	67,000	1,240
Pb	s → l	4,810	600
PbO	s → l	27,480	1,158
Si	s → l	50,200	1,685
V	s → l	22,840	2,193
Zr	α → β	3,900	1,136
ZrO ₂	α → β	5,900	1,478

Chapter 6

Solution to 6.3

(1) Change in enthalpy



In the range 298 – 716K,

$$\Delta c_p = 9.27 + 9.68 \times 10^{-3}T - 6.115 \times 10^5 T^{-2}$$

At 716 K, $\text{Ca}(\alpha) \rightarrow \text{Ca}(\beta)$, $\Delta H_{\alpha \rightarrow \beta} = 900 \text{ J}$

In the range 716 – 1000 K,

$$\Delta c_p = 35 - 38.83 \times 10^{-3}T - 6.115 \times 10^5 T^{-2}$$

$$\Delta H_{1000}$$

$$= -634900 + 9.27(716 - 298) + 4.84 \times 10^{-3}(716^2 - 298^2) + 6.115$$

$$\times 10^5 \left(\frac{1}{716} - \frac{1}{298} \right) + 900 + 35(1000 - 716) - 19.415 \times 10^{-3}$$

$$(1000^2 - 716^2) + 6.115 \times 10^5 \left(\frac{1}{1000} - \frac{1}{716} \right)$$

$$= -630836 \text{ J}$$

Chapter 6

Solution to 6.3

For $\text{Ti} + \text{O}_2 = \text{TiO}_2$, $\Delta H_{298} = -944000 \text{ J}$

In the range 298 – 1000K,

$$\Delta c_p = 23.14 - 13.47 \times 10^{-3}T - 16.53 \times 10^5 T^{-2}$$

$$\Delta H_{1000}$$

$$= -944000 + 23.14(1000 - 298) - 6.735 \times 10^{-3}(1000^2 - 298^2)$$

$$+ 16.53 \times 10^5 \left(\frac{1}{1000} - \frac{1}{298} \right)$$

$$= -937787 \text{ J}$$

Chapter 6

Solution to 6.3



In the range 298 – 716 K,

$$\Delta c_p = 35.09 - 3.78 \times 10^{-3}T - 25.485 \times 10^5 T^{-2} \text{ J/K}$$

At 716 K, $\text{Ca}(\alpha) \rightarrow \text{Ca}(\beta)$, $\Delta H_{\alpha \rightarrow \beta} = 900 \text{ J}$

In the range 716 – 1000 K, for $\text{Ca}(\beta) + \text{Ti} + \frac{3}{2}\text{O}_2 = \text{CaTiO}_3$,

$$\Delta c_p = 60.82 - 52.29 \times 10^{-3}T - 25.485 \times 10^5 T^{-2} \text{ J/K}$$

$$\begin{aligned} \Delta H_{1000} &= -1660600 + 35.09(716 - 298) - 1.89 \times 10^{-3}(716^2 - 298^2) + 25.485 \\ &\times 10^5 \left(\frac{1}{716} - \frac{1}{298} \right) + 900 + 60.82(1000 - 716) - 26.145 \times 10^{-3}(1000^2 \\ &- 716^2) + 25.485 \times 10^5 \left(\frac{1}{1000} - \frac{1}{716} \right) \\ &= -1649106 \text{ J} \end{aligned}$$

Chapter 6

Solution to 6.3

For $\text{CaO(s)} + \text{TiO}_2\text{(s)} = \text{CaTiO}_3\text{(s)}$

$$\Delta H_{1000} = -1649106 + 630836 + 937787 = -80483 \text{ J}$$

Chapter 6

Solution to 6.3

(2) Change in entropy

$$\text{For } \text{Ca}(\alpha) + \frac{1}{2} \text{O}_2 = \text{CaO}, \quad \Delta S_{298} = 38.1 - 41.6 - 205.1 \times 0.5 = -106.05 \text{ J/K}$$

$$\text{In the range } 298 - 716\text{K}, \quad \Delta c_p = 9.27 + 9.68 \times 10^{-3}T - 6.115 \times 10^5 T^{-2}$$

$$\text{in the range } 716 - 1000 \text{ K}, \quad \Delta c_p = 35 - 38.83 \times 10^{-3}T - 6.115 \times 10^5 T^{-2}$$

$$\text{At } 716 \text{ K}, \quad \text{Ca}(\alpha) \rightarrow \text{Ca}(\beta), \quad \Delta H_{\alpha \rightarrow \beta} = 900 \text{ J}$$

$$\Delta S_{1000}$$

$$= -106.05 + 9.27 \ln \left(\frac{716}{298} \right) + 9.68 \times 10^{-3} (716 - 298) + 3.0525$$

$$\times 10^5 \left(\frac{1}{716^2} - \frac{1}{298^2} \right) + \frac{900}{716} + 35 \ln \left(\frac{1000}{716} \right) - 38.83 \times 10^{-3} (1000 - 716)$$

$$+ 3.0525 \times 10^5 \left(\frac{1}{1000^2} - \frac{1}{716^2} \right) = -97.6 \text{ J/K}$$

Chapter 6

Solution to 6.3

$$\text{For Ti} + \text{O}_2 = \text{TiO}_2, \quad \Delta S_{298} = 50.6 - 205.1 - 30.7 = -185.2 \text{ J/K}$$

In the range 298 – 1000K,

$$\Delta c_p = 23.14 - 13.47 \times 10^{-3}T - 16.53 \times 10^5 T^{-2}$$

$$\begin{aligned} \Delta S_{1000} &= -185.2 + 23.14 \ln\left(\frac{1000}{298}\right) - 13.47 \times 10^{-3}(1000 - 298) + 8.265 \\ &\times 10^5 \left(\frac{1}{1000^2} - \frac{1}{298^2} \right) \\ &= -175.1 \text{ J/K} \end{aligned}$$

Chapter 6

For $\text{Ca}(\alpha) + \text{Ti} + \frac{3}{2} \text{O}_2 = \text{CaTiO}_3$,

$$\Delta S_{298} = 93.7 - 41.6 - 30.7 - \frac{3}{2} \times 205.1 = -286.25 \text{ J/K}$$

In the range 298 – 716 K,

$$\Delta c_p = 35.09 - 3.78 \times 10^{-3}T - 25.485 \times 10^5 T^{-2} \text{ J/K}$$

At 716 K, $\text{Ca}(\alpha) \rightarrow \text{Ca}(\beta)$, $\Delta H_{\alpha \rightarrow \beta} = 900 \text{ J}$

In the range 716 – 1000 K,

$$\Delta c_p = 60.82 - 52.29 \times 10^{-3}T - 25.485 \times 10^5 T^{-2} \text{ J/K}$$

ΔS_{1000}

$$\begin{aligned} &= -286.25 + 35.09 \ln\left(\frac{716}{298}\right) - 3.78 \times 10^{-3}(716 - 298) + 12.743 \\ &\times 10^5 \left(\frac{1}{716^2} - \frac{1}{298^2} \right) + \frac{900}{716} + 60.82 \ln\left(\frac{1000}{716}\right) - 52.29 \times 10^{-3}(1000 - 716) \\ &+ 12.743 \times 10^5 \left(\frac{1}{1000^2} - \frac{1}{716^2} \right) = -265.9 \text{ J/K} \end{aligned}$$

Chapter 6

Solution to 6.3

For $\text{CaO(s)} + \text{TiO}_2 = \text{CaTiO}_3\text{(s)}$

$$\Delta S_{1000} = -265.9 + 97.6 + 175.1 = 6.8 \text{ J/K}$$

6.6 Experimental verification of the third law

Assigning a value of zero to S_0 allows the absolute value of the entropy of any material to be determined as

$$S_T = \int_0^T \frac{C_P}{T} dT \text{ J/K}$$

and molar entropies are normally tabulated at 298 K, where:

$$S_{298} = \int_0^{298} \frac{C_P}{T} dT \text{ J/K}$$

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The variations, with temperature, of the molar entropies of several elements and compounds are shown in Figure 6.12. With the constant-pressure molar heat capacity of the solid expressed in the form

$$C_{P(s)} = a + bT + cT^{-2}$$

6.6 Experimental verification of the third law

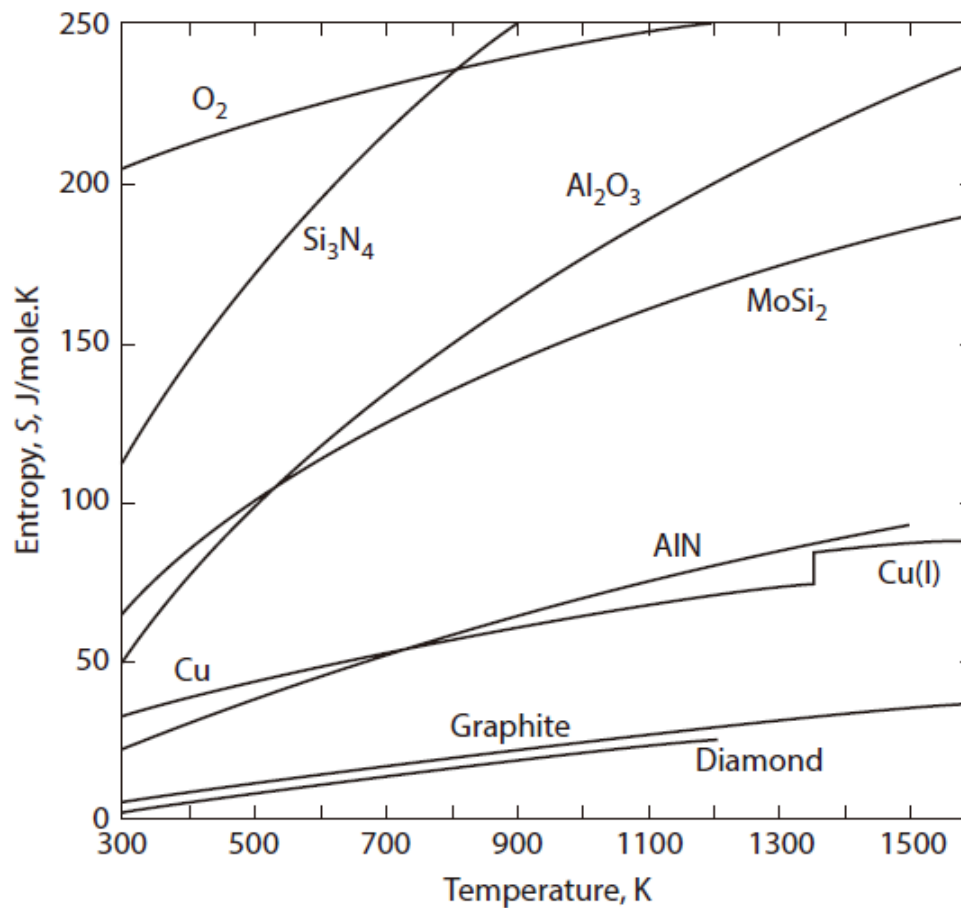


Figure 6.12 The variation, with temperature, of the molar entropies of several elements and compounds.

Chapter 6

6.4 Copper exists in the state $T=298\text{ K}$, $P=1\text{ atm}$. Calculate the temperature to which the copper must be raised at 1 atm pressure to cause the same increase in molar enthalpy as is caused by increasing its pressure to 1000 atm at 298 K.

The molar volume of Cu at 298 K is 7.09 cm^3 , and the volumetric coefficient of thermal expansion is $0.501 \times 10^{-4}\text{ K}^{-1}$. These values can be taken as being independent of pressure in the range 1– 1000 atm.

$$\Delta H = \int_{P_1}^{P_2} V(1 - \alpha T) dp.$$

6.7 The influence of pressure on enthalpy and entropy

For one mole of a closed system of fixed composition undergoing a change of pressure at constant temperature,

$$dH = \left(\frac{\partial H}{\partial P} \right)_T dP$$

$$dH = TdS + VdP \quad \Rightarrow \quad \frac{dH}{dP} = T \frac{dS}{dP} + V \quad \Rightarrow \quad \left(\frac{\partial H}{\partial P} \right)_T = T \left(\frac{\partial S}{\partial P} \right)_T + V$$

A Maxwell relation gives: $\left(\frac{\partial S}{\partial P} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_P$

$$\left(\frac{\partial H}{\partial P} \right)_T = -T \left(\frac{\partial V}{\partial T} \right)_P + V$$

6.7 The influence of pressure on enthalpy and entropy

$$\left(\frac{\partial H}{\partial P}\right)_T = -T\left(\frac{\partial V}{\partial T}\right)_P + V$$

The isobaric coefficient of thermal expansion, α , is defined as

$$\left(\frac{\partial V}{\partial T}\right)_P = \alpha V \quad \left(\frac{\partial H}{\partial P}\right)_T = -TV\alpha + V = V(1 - \alpha T)$$

The change in molar enthalpy caused by the change in state from (P_1, T) to (P_2, T) is thus

$$\Delta H = H(P_2, T) - H(P_1, T) = \int_{P_1}^{P_2} V(1 - \alpha T) dP$$

For an ideal gas, $\alpha = 1/T$, and thus, this equation shows once again that the enthalpy of an ideal gas is independent of pressure.

Chapter 6

Solution to 6.4

$$P_1=1 \text{ atm}, P_2=1000 \text{ atm}, T=298\text{K}$$

$$\begin{aligned}\Delta H &= V(1-\alpha T)(P_2 - P_1) \\ &= 7.09 \times 10^{-3} (1 - 0.493 \times 10^{-3} \times 298)(1000 - 1) \times 101.3 \\ &= 612 J\end{aligned}$$

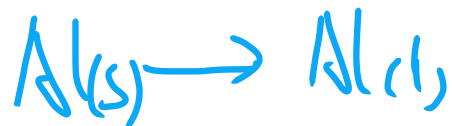
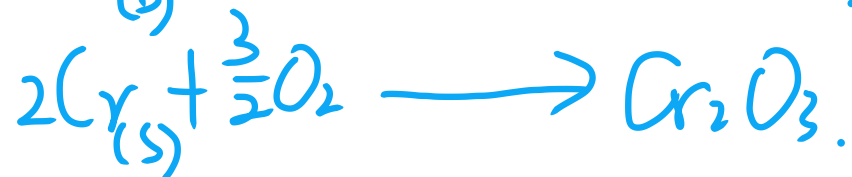
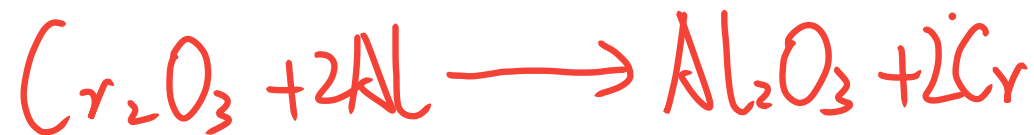
$$T_1=298\text{K}, T_2=T, P=1 \text{ atm}$$

$$\Delta H = \int_{T_1}^T c_p dT = 22.64(T - 298) - 6.28 \times 10^{-3} (T^2 - 298^2)$$

$$\text{So } T=327\text{K}$$

Chapter 6

6.6 An adiabatic vessel contains 1000 g of liquid aluminum at 700°C. Calculate the mass of Cr_2O_3 at room temperature, which, when added to the liquid aluminum (with which it reacts to form Cr and Al_2O_3), raises the temperature of the resulting mixture of Al_2O_3 , Cr_2O_3 , and Cr to 1600 K.



Chapter 6

Solution to 6.6

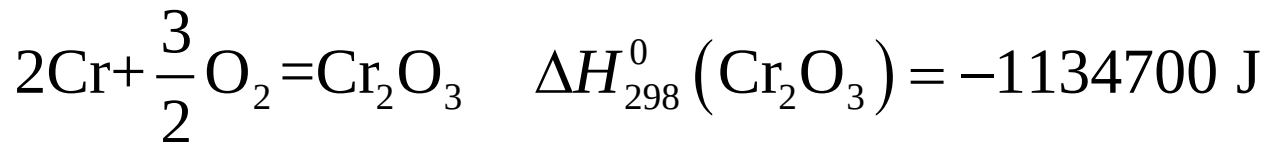
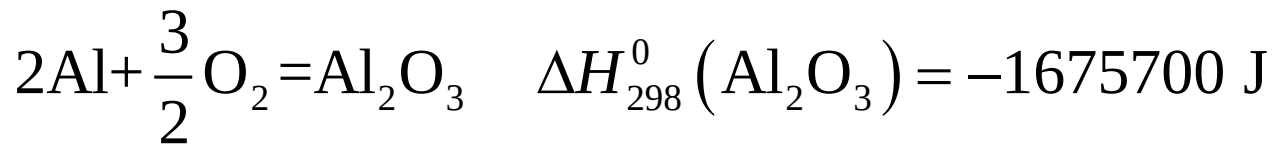
For Cr_2O_3 $c_p = 119.37 + 9.30 \times 10^{-3}T - 15.65 \times 10^5 T^{-2}$ at 298K-1800K

For Al_2O_3 $c_p = 106.6 + 17.78 \times 10^{-3}T - 28.53 \times 10^5 T^{-2}$ at 298K-2325K

For Cr(s) $c_p = 24.43 + 9.87 \times 10^{-3}T - 3.68 \times 10^5 T^{-2}$ at 298K-2130K

For Al(s) $c_p = 20.67 + 12.38 \times 10^{-3}T$ at 298K-933K

For Al(l) $c_p = 31.76$ at 933K-1600K



Chapter 6

Solution to 6.6

According to the meaning of the questions, all Al have transferred to Al_2O_3 . So in the final state, there are $500/27$ moles Al_2O_3 and $1000/27$ moles Cr and (x) moles Cr_2O_3

Process 1: the temperature of the liquid Al decrease from 973K to 298K

$$\Delta H_1 = 1000 / 27 \left(\int_{973}^{933} 31.76 dT + \Delta H(l \rightarrow s) + \int_{933}^{298} 20.67 + 12.38 \times 10^{-3} T dT \right)$$
$$= 1000 / 27 (-1270.4 - 10700 - 17964) = -1108669.635\text{J}$$

Process 2: reaction at 298K $2\text{Al} + \text{Cr}_2\text{O}_3 = 2\text{Cr} + \text{Al}_2\text{O}_3$

$$\Delta H_m^{298} = \Delta H_{298}^0 (\text{Al}_2\text{O}_3) - \Delta H_{298}^0 (\text{Cr}_2\text{O}_3) = -541000\text{J}$$

$$\Delta H_2 = \frac{500}{27} \Delta H_m^{298} = -10018218.52\text{J}$$

Chapter 6

Process 3: the temperature of the mixture raise from 298K to 1600K

$$\begin{aligned}\Delta H_3 &= 1000 / 27 \int_{298}^{1600} c_P(\text{Cr}) dT + x \int_{298}^{1600} c_P(\text{Cr}_2\text{O}_3) dT + 500 / 27 \int_{298}^{1600} c_P(\text{Al}_2\text{O}_3) dT \\&= 1000 / 27 \int_{298}^{1600} 24.43 + 9.87 \times 10^{-3} T - 3.68 \times 10^5 T^{-2} dT \\&\quad + x \int_{298}^{1600} 119.37 + 9.30 \times 10^{-3} T - 15.65 \times 10^5 T^{-2} dT \\&\quad + 500 / 27 \int_{298}^{1600} 106.6 + 17.78 \times 10^{-3} T - 28.53 \times 10^5 T^{-2} dT \\&= 1000 / 27 \left(24.43T + 4.935 \times 10^{-3} T^2 + 3.68 \times 10^5 T^{-1} \right)_{298}^{1600} \\&\quad + x \left(119.37T + 4.65 \times 10^{-3} T^2 + 15.65 \times 10^5 T^{-1} \right)_{298}^{1600} \\&\quad + 500 / 27 \left(106.6T + 8.89 \times 10^{-3} T^2 + 28.53 \times 10^5 T^{-1} \right)_{298}^{1600} \\&= 1000 / 27 \times 42998.3 + 162637.248x + 500 / 27 \times 152971.43 \\&= 4425334.404 + 162637.248x\end{aligned}$$

Chapter 6

According to the meanings of the question, we have:

$$\Delta H_1 + \Delta H_2 + \Delta H_3 = 0$$

So $x=41.2055$ mole, $n(\text{Cr}_2\text{O}_3)=x+500/27=59.72$ mole,

$m(\text{Cr}_2\text{O}_3)=n(\text{Cr}_2\text{O}_3)M(\text{Cr}_2\text{O}_3)=9078$ g

Chapter 6

6.8 Calculate the value of ΔG at 800 K for the reaction



What percentage error occurs if it is assumed that Δc_p for the reaction is zero?

Chapter 6

Solution to 6.8



Substance	$\Delta H_{298}^0(\text{J})$	$S_{298}^0(\text{J/K})$	a	b	c
Si_3N_4	-744800	113.0	70.54	98.74×10^{-3}	-
SiO_2	-910900	41.5	43.89	1.00×10^{-3}	-6.02×10^5
O_2	-	205.1	29.96	4.18×10^{-3}	-1.67×10^5
N_2	-	191.5	27.87	4.27×10^{-3}	-

$$c_p = a + bT + cT^{-2} \quad (\text{J/mole} \cdot \text{K})$$

Chapter 6



$$\begin{aligned}\Delta H_{298} &= 3\Delta H_{\text{SiO}_2} + 2\Delta H_{\text{N}_2} - (\Delta H_{\text{Si}_3\text{N}_4} + 3\Delta H_{\text{O}_2}) \\ &= 3 \times (-910900) - (-744800) \\ &= -1987900 \text{ J}\end{aligned}$$

$$\begin{aligned}\Delta S_{298} &= 3S_{\text{SiO}_2} + 2S_{\text{N}_2} - (S_{\text{Si}_3\text{N}_4} + 3S_{\text{O}_2}) \\ &= 3 \times 41.5 + 2 \times 191.5 - (113.0 + 3 \times 205.1) \\ &= -220.8 \text{ J/K}\end{aligned}$$

$$\begin{aligned}\Delta c_p &= 3 \times 43.89 + 3 \times 1.0 \times 10^{-3}T - 3 \times 6.02 \times 10^5 T^{-2} + 2 \times 27.87 + 2 \\ &\quad \times 4.27 \times 10^{-3}T - 70.54 - 98.74 \times 10^{-3}T - 3 \times 29.96 - 3 \times 4.18 \\ &\quad \times 10^{-3}T + 3 \times 1.67 \times 10 T^{-2} \\ &= 26.99 - 0.09974T - 1305000.0T^{-2}\end{aligned}$$

Chapter 6

$$\begin{aligned}\Delta H_{800} &= \Delta H_{298} + \int_{298}^{800} \Delta c_p dT \\ &= -1987900 + \int_{298}^{800} (26.99 - 0.09974T - 1305000.0T^{-2}) dT \\ &= -2004587.1 \text{ J}\end{aligned}$$

$$\begin{aligned}\Delta S_{800} &= \Delta S_{298} + \int_{298}^{800} \frac{\Delta c_p}{T} dT \\ &= -220.8 + \int_{298}^{800} \frac{(26.99 - 0.09974T - 1305000.0T^{-2})}{T} dT \\ &= -250.5 \text{ J/K}\end{aligned}$$

$$\begin{aligned}\Delta G_{800} &= \Delta H_{800} - 800\Delta S_{800} \\ &= -2004587.1 - 800 \times (-250.5) \\ &= -1804.2 \text{ kJ}\end{aligned}$$

Chapter 6

If $\Delta c_p = 0$, then

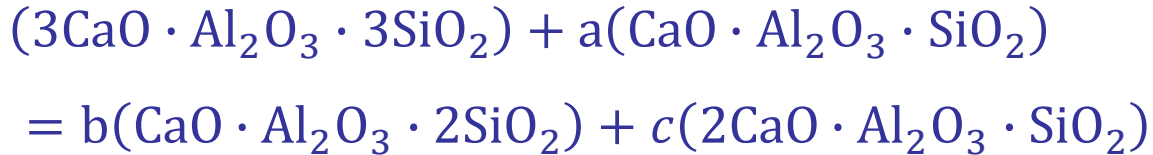
$$\begin{aligned}\Delta G_{800} &= \Delta H_{298} - 800\Delta S_{298} \\ &= -1987900 - 800 \times (-220.8) \\ &= -1811.3 \text{ kJ}\end{aligned}$$

$$Error\% = \frac{1811.3 - 1804.2}{1804.2} \times 100\% = 0.39\%$$

The error is only about 0.4%

Chapter 6

6.9 Determine the stoichiometric coefficients for the reaction



and calculate ΔH_{298} , ΔS_{298} , and ΔG_{298} for the reaction.

Substance	$\Delta H_{298}^0(\text{J})$	$S_{298}^0(\text{J/K})$
$3\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2$	– 6,646,300	241.4
$\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	– 3,293,200	144.8
$\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$	– 4,223,700	202.5
$2\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	– 3,989,400	198.3

Chapter 6

Solution to 6.9

$$(3\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2) + a(\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2) \\ = b(\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2) + c(2\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2)$$

According to the conservation of mass

$$3 + a = b + 2c$$

$$1 + a = b + c$$

$$3 + a = 2b + c$$

$$a = 3$$

$$b = c = 2$$

So

Substance	$\Delta H_{298}^0(\text{J})$
$3\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2$	- 6,646,300
$\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	- 3,293,200
$\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$	- 4,223,700
$2\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	- 3,989,400

$$\Delta H_{298}$$

$$= -\Delta H_{298}(3\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2) - 3\Delta H_{298}(\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2)$$

$$+ 2\Delta H_{298}(\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2) + 2\Delta H_{298}(2\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2)$$

$$= 6646300 + 3 \times 3293200 - 2 \times 4223700 - 2 \times 3989400$$

$$= 99700\text{J}$$

Chapter 6

$$\Delta S_{298}$$

$$= -S_{298}(3\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2) - 3S_{298}(\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2)$$

$$+ 2S_{298}(\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2) + 2S_{298}(2\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2)$$

$$= -241.4 - 3 \times 144.8 + 2 \times 202.5 + 2 \times 198.3$$

$$= 125.8 \text{ J/K}$$

$$\Delta G_{298} = \Delta H_{298} - T\Delta S_{298} = 62211.6 \text{ J}$$

Substance	$S_{298}^0(\text{J/K})$
$3\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2$	241.4
$\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	144.8
$\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$	202.5
$2\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	198.3

Chapter 6

6.11 *A function $\Omega \equiv -A + \mu N$, has been defined, where A is the Helmholtz free energy, μ is the chemical potential and N is the number of particles in the system. Obtain an expression for $d\Omega$ and thus determine the independent variables for this function.

Chapter 6

Solution to 6.11

$$d\Omega = -dA + \mu dN + Nd\mu$$

$$dA = -SdT - PdV + \mu dN$$

$$d\Omega = SdT + PdV - \mu dN + \mu dN + Nd\mu$$

$$d\Omega = SdT + PdV + Nd\mu$$

$$\Omega = \Omega(T, V, \mu)$$

Chapter 6

6.12*(a) Given that the Helmholtz energy can be written as

$$A = -N_{\text{system}} k_B T \ln Z$$

Obtain simplified expressions for both A and S of an Einstein solid.

(b) Show that the entropy approaches 0 as the temperature approaches 0 in such a solid.

Chapter 6

Solution to 6.12

$$\varepsilon_i = \left(i + \frac{1}{2}\right) h\nu$$

From the text we saw that Z for the Einstein solid can be written as:

$$Z = \sum \exp\left(-\frac{\varepsilon_i}{kT}\right) = \exp\left[-\frac{1}{2} \frac{h\nu}{k_B T}\right] \left[\frac{1}{1 - \exp\left(\frac{-h\nu}{k_B T}\right)} \right]$$

$$A = -N_{\text{system}} k_B T \ln Z$$

$$A = \frac{1}{2} N_{\text{system}} h\nu + N_{\text{system}} k_B T \ln \left[1 - \exp\left(\frac{-h\nu}{k_B T}\right) \right]$$

$$-\left(\frac{\partial A}{\partial T}\right)_V = S = N_{\text{system}} k_B \ln Z + N_{\text{system}} k_B T \left(\frac{\partial \ln Z}{\partial T}\right)_V$$

Chapter 6

$$S = N_{\text{system}} k_B \ln Z + N_{\text{system}} k_B T \left(\frac{\partial \ln Z}{\partial T} \right)_V$$

$$Z = \exp \left[-\frac{1}{2} \frac{h\nu}{k_B T} \right] \left[\frac{1}{1 - \exp \left(\frac{-h\nu}{k_B T} \right)} \right]$$

$$S = N_{\text{system}} k_B \left(\frac{h\nu}{k_B T} \right) \frac{\exp \left(\frac{-h\nu}{k_B T} \right)}{1 - \exp \left(\frac{-h\nu}{k_B T} \right)} - N_{\text{system}} k_B \ln \left[1 - \exp \left(\frac{-h\nu}{k_B T} \right) \right]$$

Since $N_{\text{system}} = 3n$

$$S = 3n k_B \left(\frac{h\nu}{k_B T} \right) \frac{\exp \left(\frac{-h\nu}{k_B T} \right)}{1 - \exp \left(\frac{-h\nu}{k_B T} \right)} - 3n k_B \ln \left[1 - \exp \left(\frac{-h\nu}{k_B T} \right) \right]$$

Chapter 6

$$S = 3nk \left(\frac{h\nu}{k_B T} \right) \frac{\exp\left(\frac{-h\nu}{k_B T}\right)}{1 - \exp\left(\frac{-h\nu}{k_B T}\right)} - 3nk_B \ln \left[1 - \exp\left(\frac{-h\nu}{k_B T}\right) \right]$$

$$\frac{\left(\frac{h\nu}{k_B T} \right) \exp\left(\frac{-h\nu}{k_B T}\right)}{1 - \exp\left(\frac{-h\nu}{k_B T}\right)} \rightarrow 0 \text{ as } T \rightarrow 0$$

$$1 - \exp\left(\frac{-h\nu}{k_B T}\right) \rightarrow 1 \text{ as } T \rightarrow 0; \ln 1 = 0$$

So $S \rightarrow 0$ as $T \rightarrow 0$

Chapter 6

6.13* Obtain an expression for the internal energy of an Einstein solid as the temperature gets very large, differentiate it with respect to T , and show that the Dulong and Petit value for the heat capacity is attained.

Chapter 6

Solution to 6.13

$$U' = \frac{3}{2}nh\nu + \frac{3nh\nu}{\left(\exp\left(\frac{h\nu}{k_B T}\right) - 1\right)}$$

$$\text{as } T \rightarrow \infty \quad \exp\left(\frac{h\nu}{k_B T}\right) \rightarrow 1 + \frac{h\nu}{k_B T}$$

$$U' = \frac{3}{2}nh\nu + \frac{3nh\nu}{\left(\frac{h\nu}{k_B T}\right)} = \frac{3}{2}nh\nu + 3nk_B T$$

$$\text{Thus } C_V = \frac{\partial U'}{\partial T} = 3nk_B = 3R \quad \text{for one mole substance}$$

Summary of Chapter 7

1. Knowledge of the dependencies, on temperature and pressure, of the changes in molar enthalpy and molar entropy caused by phase changes in a system allows determination of the corresponding change in the molar Gibbs free energy of the system.
2. Since a closed one-component system has only two independent variables, the dependence of G can be examined most simply by choosing the state variables T and P as the independent variables (these are the natural independent variables when G is the dependent variable). The phases in which the material can exist can thus be represented in a three-dimensional diagram using the state function G , the independent state variables P , and T as coordinates.

Summary of Chapter 7

3. If sections of these three-dimensional diagrams are taken at constant P , one obtains G versus T plots for the phases. The phase that is in equilibrium at a given temperature is the one with the lowest G . Crossings of the curve represent temperatures at which two (or three) phases are in equilibrium. The slope of the G versus T plot is the negative of the entropy, and the curvature is related to the heat capacity of the phases delineated by the curves.

4. Sections of the three-dimensional diagrams may be taken at constant temperature. Such plots of P versus T show which phase is in equilibrium as a function of pressure. Their slopes are equal to the molar volume of the phases, and their curvatures are related to the isothermal compressibility of the phase being plotted.

Summary of Chapter 7

5. In the three-dimensional diagrams, the various states in which the material can exist occur as surfaces. In any state, which is determined by the values of P and T , the stable phase is that which has the lowest Gibbs free energy. The surfaces in the diagram intersect with one another along curves, and these curves represent the variations of P with T required for equilibrium between the two phases.

Summary of Chapter 7

6. The intersection of the surfaces for the solid and liquid phases gives the variation of the equilibrium melting temperature with pressure. The intersection of the surfaces for the liquid and vapor phases gives the variation of the boiling temperature with pressure. The normal melting and boiling points of the material occur on these intersections at $P = 1$ atm. Three surfaces intersect at a point in the diagram, and the values of P and T at which this intersection occurs are those of the invariant triple point at which an equilibrium occurs among three phases. In a one-component system, no more than three phases can coexist in equilibrium with one another.

Summary of Chapter 7

7. The three-dimensional G - T - P diagram illustrates the differences between stable, metastable, and unstable states and, hence, shows the difference between reversible and irreversible process paths. At any value of P and T , the stable phase is that which has the lowest Gibbs free energy, and phases which have higher values of G at the same values of P and T are metastable with respect to the phase of lowest value of G . Phases with a value of G at any combination of P and T which do not lie on a surface in the diagram are unstable. A reversible process path involving a change in P and/or T lies on a phase surface, and the state of a phase is changed reversibly only when, during the change, the state of the system does not leave the surface of the phase. If the process path leaves the phase surface, then the change of state, which necessarily passes through non-equilibrium states, is irreversible.

Summary of Chapter 7

8. Since the perspective representation, in two dimensions, of a three-dimensional diagram is difficult, it is normal practice to present the phase diagram for a one component system as the basal plane of the G - T - P diagram (i.e., a P - T diagram), onto which are projected the lines along which two surfaces intersect (equilibrium between two phases) and the points at which three surfaces intersect (equilibrium among three phases). Such a diagram contains areas in which a single phase is stable, which are separated by curves along which two phases exist at equilibrium, and points at the intersection of three curves at which three phases coexist in equilibrium. The curves for equilibrium between a condensed phase and the vapor phase are called *vapor pressure curves*, and they are exponential in form. In view of the fact that saturated vapor pressures can vary over several orders of magnitude, pressure–temperature phase diagrams can often be presented in more useful form as plots of $\ln P$ versus $1/T$ than as plots of P versus T .

Summary of Chapter 7

- 9.** The development of phase diagrams for one-component systems demonstrates the use of the Gibbs free energy as a criterion for equilibrium when T and P are chosen as the independent state variables.
- 10.** The equilibrium Gibbs phase rule for a one-component simple system is where F is the number of degrees of freedom and the ϕ number of phases allowed to be present in equilibrium.
- $$F = 3 - \phi$$
- 11.** Thermodynamic equilibrium is affected by the application of external fields, such as a magnetic or electrical field.

Chapter 7

7.1 Using the vapor pressure–temperature relationships for

$\text{CaF}_2(\alpha)$, $\text{CaF}_2(\beta)$, and liquid CaF_2 , calculate

a. The temperatures and pressures of the triple points for the equilibria

$\text{CaF}_2(\alpha) - \text{CaF}_2(\beta) - \text{CaF}_2(v)$ and $\text{CaF}_2(\beta) - \text{CaF}_2(l) - \text{CaF}_2(v)$

b. The normal boiling temperature of CaF_2

c. The molar latent enthalpy of the transformation $\text{CaF}_2(\alpha) \rightarrow \text{CaF}_2(\beta)$

d. The molar latent enthalpy of melting of $\text{CaF}_2(\beta)$

Chapter 7

Solution to 7.1

The saturated vapor pressures of the three state of CaF_2 are:

$$\text{CaF}_2(\alpha): \ln p \text{ (atm)} = -54350/T - 4.525 \ln T + 56.57$$

$$\text{CaF}_2(\beta): \ln p \text{ (atm)} = -53780/T - 4.525 \ln T + 56.08$$

$$\text{CaF}_2(l): \ln p \text{ (atm)} = -50200/T - 4.525 \ln T + 53.96$$

a. The temperatures and pressures of the triple points for the equilibria

$\text{CaF}_2(\alpha) - \text{CaF}_2(\beta) - \text{CaF}_2(v)$:

$$-54350/T - 4.525 \ln T + 56.57 = -53780/T - 4.525 \ln T + 56.08 \Rightarrow$$

$$T = 1163.26 \text{ K}$$

$$p = 2.54 \times 10^{-10} \text{ Pa}$$

Chapter 7

$$\text{CaF}_2(\beta): \ln p \text{ (atm)} = -53780/T - 4.525 \ln T + 56.08$$

$$\text{CaF}_2(l): \ln p \text{ (atm)} = -50200/T - 4.525 \ln T + 53.96$$

The temperatures and pressures of the triple points for the equilibria

$\text{CaF}_2(\beta) - \text{CaF}_2(l) - \text{CaF}_2(v)$:

$$-\frac{53780}{T} - 4.525 \ln T + 56.08 = -\frac{50200}{T} - 4.525 \ln T + 53.96 \Rightarrow$$

$$T = 1688.67 \text{ K}$$

$$p = 8.31 \times 10^{-5} \text{ atm}$$

Chapter 7

b. The normal boiling temperature of CaF_2

When $p=1$ atm, we have

$$0 = \frac{50200}{T_b} - 4.525 \ln T_b + 53.96$$

$$\Rightarrow T_b = 2775.71 \text{ K}$$

$$\text{CaF}_{2(l)}: \ln p (\text{atm}) = -50200/T - 4.525 \ln T + 53.96$$

Chapter 7

c. The molar latent enthalpy of the transformation $\text{CaF}_2(\alpha) \rightarrow \text{CaF}_2(\beta)$

According to the Clausius-Clapeyron equation $d \ln p = \frac{\Delta H}{RT^2} dT$, we have:

$$\frac{d \left(-\frac{54350}{T} - 4.525 \ln T + 56.57 \right)}{dT} = \frac{\Delta H_{(\alpha \rightarrow v)}}{RT^2}$$

$$\frac{d \left(-\frac{53780}{T} - 4.525 \ln T + 56.08 \right)}{dT} = \frac{\Delta H_{(\beta \rightarrow v)}}{RT^2}$$

$$\frac{d \left(-\frac{50200}{T} - 4.525 \ln T + 53.96 \right)}{dT} = \frac{\Delta H_{(l \rightarrow v)}}{RT^2}$$

Chapter 7

$$\Delta H_{(\alpha \rightarrow v)} = 54350R - 4.525RT$$

$$\Delta H_{(\beta \rightarrow v)} = 53780R - 4.525RT$$

$$\Delta H_{(l \rightarrow v)} = 50200R - 4.525RT$$

So,

$$\Delta H_{(\alpha \rightarrow \beta)} = \Delta H_{(\alpha \rightarrow v)} - \Delta H_{(\beta \rightarrow v)} = 570R = 4739.265 \text{ J}$$

d. The molar latent enthalpy of melting of $\text{CaF}_2(\beta)$

We can also get,

$$\Delta H_{(\beta \rightarrow l)} = \Delta H_{(\beta \rightarrow v)} - \Delta H_{(l \rightarrow v)} = 3580R = 29765.91 \text{ J}$$

7.5 Equilibrium between the vapor phase and a condensed phase

For condensed phase–vapor equilibria, the Clapeyron equation can be written as

$$\left(\frac{dP}{dT}\right)_{eq} = \frac{\Delta H}{T \Delta V} \xrightarrow{\Delta V = V^{\text{vapor}}} \left(\frac{dP}{dT}\right)_{eq} = \frac{\Delta H}{TV^{\text{vapor}}}$$

If it is further assumed that the vapor in equilibrium with the condensed phase behaves ideally (i.e., $PV=RT$), then

$$\left(\frac{dP}{dT}\right)_{eq} = \frac{P \Delta H}{RT^2}$$

7.5 Equilibrium between the vapor phase and a condensed phase

$$\left(\frac{dP}{dT}\right)_{eq} = \frac{P\Delta H}{RT^2} \quad \Rightarrow \quad \frac{dP}{P} = \frac{\Delta H}{RT^2} dT$$



$$d \ln P = \frac{\Delta H}{RT^2} dT$$

This is Clausius–Clapeyron equation

7.5 Equilibrium between the vapor phase and a condensed phase

$$d \ln P = \frac{\Delta H}{RT^2} dT$$

If ΔH is independent of temperature — that is, if $c_p(\text{vapor})=c_p(\text{condensed phase})$ — integration this equation gives

$$\ln P = \frac{\Delta H}{RT} + \text{constant}$$

where the initial pressure is assumed to be 1 atm.

If we use the boiling point values of the pressure and temperature (P_0 and T_b), we can write

$$\ln \frac{P}{P_0} = -\frac{\Delta H}{R} \left(\frac{1}{T} - \frac{1}{T_b} \right)$$

Chapter 7

7.3 One mole of SiCl_4 vapor is contained at 1 atm pressure and 350 K in a rigid container of fixed volume. The temperature of the container and its contents is cooled to 280 K. At what temperature does condensation of the SiCl_4 vapor begin, and what fraction of the vapor has condensed when the temperature is 280 K?

Chapter 7

Solution to 7.3

The saturated vapor pressures of the SiCl_4 is

$$\ln p = -3620 / T + 10.96 \Rightarrow \ln V = 3620 / T - 10.96 + \ln RT$$

At first, there is one mole of SiCl_4 vapor contained at 1 atm pressure and 350 K in a rigid container of fixed volume. So $V=2910.75 \text{ J/atm}$

$$\ln 2910.75 = 3620 / T - 10.96 + \ln 8.3145T \Rightarrow T = 328.3701 \text{ K}$$

Chapter 7

When $T=280\text{K}$, the system reaches equilibrium. so

$$\ln p = -3620 / 280 + 10.96 \Rightarrow P = 0.1396 \text{ atm}$$

$$PV = nRT \Rightarrow n_{350} = \frac{V}{350R}, n_{280} = \frac{0.1396V}{280R}$$

$$\eta = \frac{\frac{V}{350R} - \frac{0.1396V}{280R}}{\frac{V}{350R}} = 82.55\%$$

Chapter 7

7.6 Below the triple point ($-56.2\text{ }^{\circ}\text{C}$), the vapor pressure of solid CO_2 is given as

$$\ln P(\text{atm}) = -\frac{3116}{T} + 16.01$$

The molar latent enthalpy of melting of CO_2 is 8330 J . Calculate the vapor pressure exerted by liquid CO_2 at $25\text{ }^{\circ}\text{C}$ and explain why solid CO_2 is referred to as “dry ice”

Chapter 7

Solution to 7.6

– 56.2 °C

$$\ln P(\text{atm}) = -\frac{3116}{T} + 16.01$$

$$d \ln P = \frac{\Delta H}{RT^2} dT$$

The triple point pressure is

$$\ln p = -3116 / 216.8 + 16.01 \Rightarrow p = 5.14 \text{ atm}$$

$$\frac{d \ln p}{dT} = \frac{3116}{T^2} = \frac{\Delta H_{(s \rightarrow v)}}{RT^2}$$

$$\Delta H(s \rightarrow v) = 8.3144 \times 3116 = 25907 \text{ J}$$

$$\Delta H(s \rightarrow l) = 8330 \text{ J}$$

$$\Delta H(l \rightarrow v) = 25907 - 8330 = 17577 \text{ J}$$

Chapter 7

For liquid CO₂

$$\ln p_{\text{CO}_2(l)} = \frac{-\Delta H_{(l \rightarrow v)}}{RT} + \text{constant}$$

At the triple point

$$\ln 5.14 = \frac{-17577}{216.8R} + \text{constant} \Rightarrow \text{constant} = 11.39$$

$$\ln p_{\text{CO}_2(l)} = \frac{-\Delta H_{(l \rightarrow v)}}{RT} + 11.39$$

At 298K

$$p_{\text{CO}_2(l)} = 73.3 \text{ atm}$$

The triple-point pressure is 5.14 atm, and, as the 1 atm isobar does not pass through the liquid-phase field, liquid CO₂ is not stable at atmospheric pressure at 25 °C.

Chapter 7

7.7 The molar volumes of solid and liquid lead (铅) at the normal melting temperature of lead are, respectively, 18.92 and 19.47 cm³. Calculate the pressure which must be applied to lead in order to increase its melting temperature by 20 °C.

Chapter 7

Solution to 7.7

For Pb at 600K

$$\Delta H (s \rightarrow l) = 4810 \text{ J}$$

$$\left(\frac{dp}{dT} \right)_{eq} = \frac{\Delta H}{T \Delta V} = \frac{4810}{T (19.47 - 18.92) \times 10^{-6}} = \frac{8.745 \times 10^9}{T}$$

Increase its melting temperature by 20 °C

$$P - 1 = \int_{600}^{620} \frac{8.745 \times 10^9}{T} dT = 8.745 \times 10^9 \ln T_{600}^{620} = 2829 \text{ atm}$$

So $P=2830 \text{ atm}$

Chapter 7

7.9 Measurements of the saturated vapor pressure of liquid NdCl_5 give 0.3045 atm at 478 K and 0.9310 atm at 520 K. Calculate the normal boiling temperature of NdCl_5 .

Chapter 7

Solution to 7.9

The vapor pressure equation is of the form

0.3045 atm at 478 K
0.9310 atm at 520 K

$$\ln p = \frac{A}{T} + B$$

$$\ln(0.3045) = \frac{A}{478} + B$$

$$\ln(0.9310) = \frac{A}{520} + B$$

So, $A = -6614$ and $B = 12.56$

At $P = 1 \text{ atm}$, $T_b = 523 \text{ K}$

Chapter 7

7.10* Three allotropes α , β and γ of a certain element are in equilibrium at its triple point (Figure 7.19). It is known that

$$V_m^\gamma > V_m^\alpha \text{ and } S_m^\gamma < S_m^\beta$$

Determine which regions of the diagram are α , β and γ . Explain your reasoning.

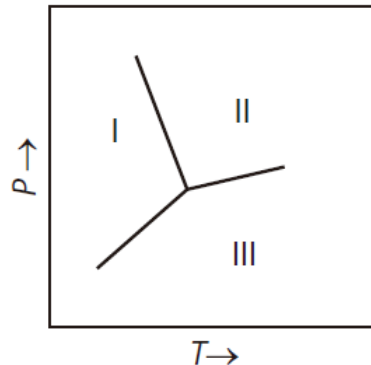


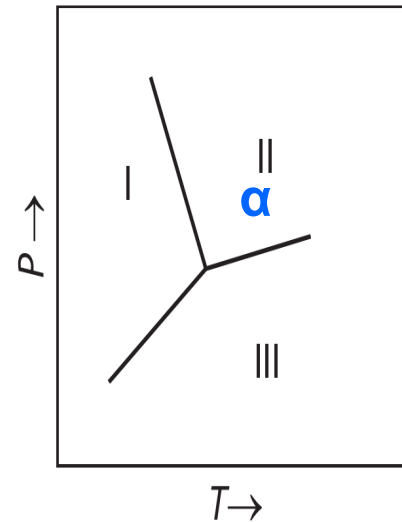
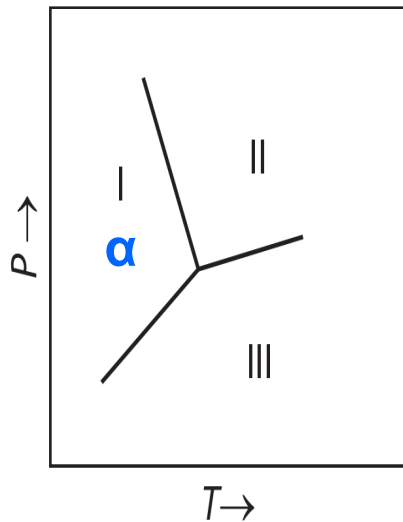
Figure 7.19 Schematic pressure–temperature phase diagram for a one-component system in the vicinity of its triple point. All phases are solids.

Chapter 7

Solution to 7.10

If the pressure is increased, equilibrium of the system favors the lowest molar volume phase.

Therefore, α is in either region I or II of Figure 7.19, since $V_m^\gamma > V_m^\alpha$.



Chapter 7

Assumed I= α , if II= β , then
III= γ .

$$\left(\frac{dP}{dT}\right)_{\alpha\gamma} = \frac{\Delta S}{\Delta V} = \frac{S^\gamma - S^\alpha}{V^\gamma - V^\alpha} > 0$$

$$\therefore V_m^\gamma > V_m^\alpha$$

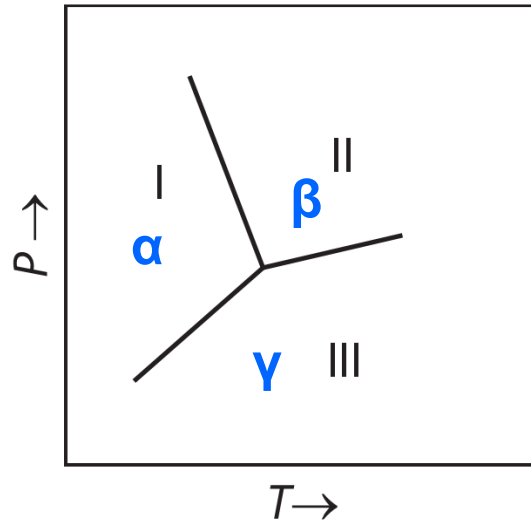
$$\therefore S^\gamma > S^\alpha$$

$$\left(\frac{dP}{dT}\right)_{\beta\gamma} = \frac{\Delta S}{\Delta V} = \frac{S^\gamma - S^\beta}{V^\gamma - V^\beta} > 0$$

$$\therefore S^\beta > S^\gamma$$

$$\therefore V_m^\beta > V_m^\gamma$$

$$\therefore V_m^\beta > V_m^\gamma > V_m^\alpha$$



$$\left(\frac{dP}{dT}\right)_{\alpha\beta} = \frac{\Delta S}{\Delta V} = \frac{S^\beta - S^\alpha}{V^\beta - V^\alpha} < 0$$

$$\therefore V_m^\beta > V_m^\alpha,$$

$$\therefore S^\alpha > S^\beta$$

$$\therefore S^\beta > S^\gamma$$

$$\therefore S^\alpha > S^\beta > S^\gamma$$

$$\therefore S^\alpha > S^\gamma$$

It is impossible since $S^\gamma > S^\alpha$.

$$\therefore \beta = \text{III}, \gamma = \text{II}.$$

Chapter 7

But, at the triple point, increasing the pressure moves equilibrium to the lowest molar volume phase.

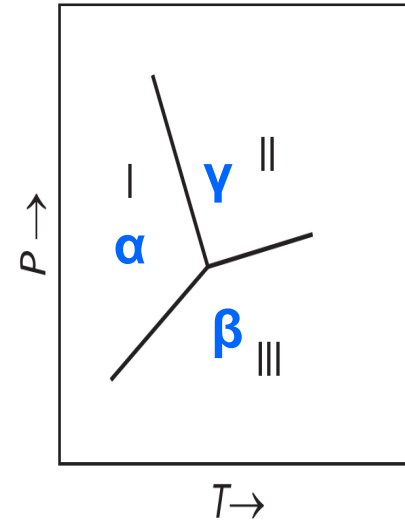
$$\because V_m^\gamma > V_m^\alpha$$

$$\therefore \gamma \neq \text{II}$$

$$\therefore \alpha \neq \text{I}$$

$$\therefore \alpha = \text{II}$$

Thus, region II is the α phase field.



Chapter 7

If I=β, then III=γ.

$$\left(\frac{dP}{dT}\right)_{\beta\gamma} = \frac{\Delta S}{\Delta V} = \frac{S^\gamma - S^\beta}{V^\gamma - V^\beta} > 0$$

$$\therefore S^\beta > S^\gamma$$

$$\therefore V_m^\beta > V_m^\gamma$$

$$\therefore V_m^\beta > V_m^\gamma > V_m^\alpha$$

$$\left(\frac{dP}{dT}\right)_{\alpha\gamma} = \frac{\Delta S}{\Delta V} = \frac{S^\gamma - S^\alpha}{V^\gamma - V^\alpha} > 0$$

$$\therefore V_m^\gamma > V_m^\alpha$$

$$\therefore S^\gamma > S^\alpha$$

$$\therefore S^\beta > S^\gamma$$

$$\therefore S^\beta > S^\gamma > S^\alpha$$

$$\left(\frac{dP}{dT}\right)_{\beta\alpha} = \frac{\Delta S}{\Delta V} = \frac{S^\alpha - S^\beta}{V^\alpha - V^\beta} < 0$$

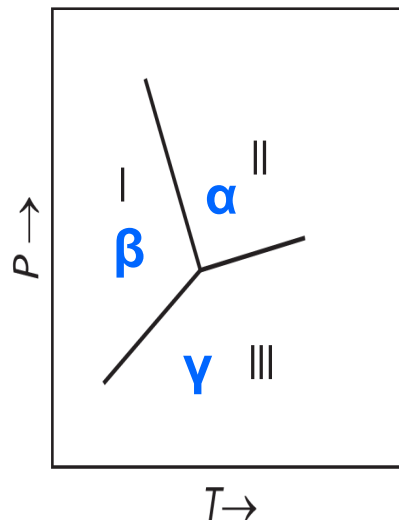
$$\therefore S^\beta > S^\alpha$$

$$\therefore V_m^\beta < V_m^\alpha$$

It is impossible since $V_m^\beta > V_m^\gamma > V_m^\alpha$.

$$\therefore \beta \neq I$$

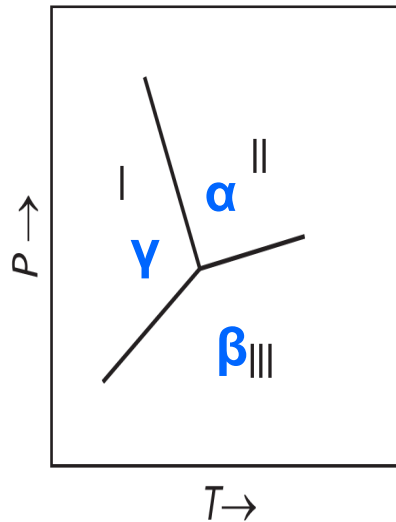
$$\therefore \beta = III, \gamma = I.$$



Chapter 7

Solution to 7.10

Therefore



$$V_m^\gamma > V_m^\alpha \text{ and } S_m^\gamma < S_m^\beta$$

Thermodynamics & Phase transformations

The End

